CSE331	Automata and Computability

Lecture 3.1: Defining Regular Expressions

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In this lecture, you will learn the formal definition and the properties of regular expression.

We learnt from the previous lecture that-

- Informally, any expression that you can create using OR (|), Concatenation (.) and Closure (*) operators are called regular expression.
- Any string that matches the regular expression must be a member of the associated regular language.
- Every string of the regular language must match with the regular expression.

For example, the expression $(a|b)^*ab$ | ϵ | $baab$ is regular based on the informal definition described above as it contains some symbols (a, b, ϵ) along with only OR, concatenation and closure operators. But this definition is somewhat vague as well as not formal.

Formal Definition of Regular Expression

The definition of Regular Expression is **Inductive**. That means we need to establish a couple of **base statements**, and then define the whole class of regular expression by ***growing*** the definition based on those statements.

1. **Base statements:**
    - The empty set $\boldsymbol{\phi}$ is a regular expression; its language is $\{\}$
    - $\boldsymbol{\epsilon}$ itself is a regular expression; its language is $\{\epsilon\}$
    - Any symbol $\boldsymbol{a}$ from the alphabet $\boldsymbol{\Sigma}$ is a regular expression; its language is $\{a\}$
2. **The Inductive Definition:** If $Rex1$ and $Rex2$ are two regular expressions, then –
    - $\boldsymbol{Rex1|Rex2}$ is a regular expression
    - $\boldsymbol{Rex1 \cdot Rex2}$ is a regular expression

    Again, if $\boldsymbol{Rex}$ is a regular expression, then –
    - $\boldsymbol{Rex^*}$ is a regular expression
    - $(\boldsymbol{Rex})$ is a regular expression (i.e. Putting brackets does not change a regular expression.)

Question 1

$(\boldsymbol{a}|\boldsymbol{b})^*\boldsymbol{ab}$ **|** $\boldsymbol{\epsilon}$ **|** $\boldsymbol{baab}$ - Is this a regular expression?

Answer 1

Observe that it has three parts separated by OR (|). $(a|b)^*ab$; ϵ; and $baab$.

- **Base statements:**
  - $\boldsymbol{\epsilon}$ is a regular expression.
  - $\boldsymbol{a}$ is a regular expression
  - $\boldsymbol{b}$ is a regular expression
- **Part 1 proof**: $(\boldsymbol{a}|\boldsymbol{b})^*\boldsymbol{ab}$

  $\boldsymbol{a}$ is a regular expression. (base case)
  $\boldsymbol{b}$ is a regular expression. (base case)
  ⇒ $\boldsymbol{a}|\boldsymbol{b}$ is a regular expression. (OR)
  ⇒ $(\boldsymbol{a}|\boldsymbol{b})$ is a regular expression. (brackets cannot modify a regular expression)
  ⇒ $(\boldsymbol{a}|\boldsymbol{b})^*$ is a regular expression. ($Kleene\ Closure$)
  ⇒ $(\boldsymbol{a}|\boldsymbol{b})^*\boldsymbol{a}$ is a regular expression. ($Concatenation$)
  ⇒ $(\boldsymbol{a}|\boldsymbol{b})^*\boldsymbol{ab}$ is a regular expression. ($Concatenation$)
- **Part 2 proof:** $\boldsymbol{\epsilon}$
  ϵ is a regular expression (base case)
- **Part 3 proof:** $\boldsymbol{baab}$
  $\boldsymbol{b}$ is a regular expression. (base case)
  $\boldsymbol{a}$ is a regular expression. (base case)
  ⇒ $\boldsymbol{ba}$ is a regular expression. ($Concatenation$)
  ⇒ $\boldsymbol{baa}$ is a regular expression. ($Concatenation$)
  ⇒ $\boldsymbol{baab}$ is a regular expression. ($Concatenation$)
- **Finally:** $(\boldsymbol{a}|\boldsymbol{b})^*\boldsymbol{ab}$ | $\boldsymbol{\epsilon}$ | $\boldsymbol{baab}$ (Part 1, Part 2 and Part 3 are ORed together)

So, $(a|b)^*ab$ | ϵ | $baab$ is a regular expression.

Question 2

$\boldsymbol{a^2b^4}|\boldsymbol{b^2a^4}$ - Is this a regular expression?

Answer 2

This is basically a short form of $aabbbb$ | $bbaaaa$. As Only $concatenation$ as well as OR are present, from the discussion above, this is a RegEx.

Question 3

$\boldsymbol{a^m b^n c^k}$- Is this a regular expression where $\boldsymbol{m}, \boldsymbol{k} \geq \mathbf{0}, \boldsymbol{n} \geq \mathbf{1}$?

Answer 3

If we check the conditions of $\boldsymbol{m}, \boldsymbol{n}$ & $\boldsymbol{k}$**,** we shall understand that this expression is nothing but this- $\boldsymbol{a}^*\boldsymbol{b} + \boldsymbol{c}^*$. Now, this is obvious that the above one is a RegEx.

Question 4

$\boldsymbol{a^m b^{2m}}$- Is this a regular expression where $\boldsymbol{m} \geq \boldsymbol{0}$?

Answer 4

This one is not regular. Because, there is no way to ensure the relationship between the number of $\boldsymbol{a}$ and $\boldsymbol{b}$**.** This just say that the number of $\boldsymbol{b}$ should be preceded by exactly twice the number of $\boldsymbol{a}$. This kind of relation is not possible in RegEx.

Shorthand Notations

Besides the above three basic operators, there are some shorthand notations widely used in RegEx.

- **? notation:** ? means **zero or one occurrence.** So, $r?$ simply means 0 or 1 occurrence of $\boldsymbol{r}$ where r can be a symbol or a RegEx. In short, $\boldsymbol{r? = r|\epsilon}$**.**
  For example- $ab?c$ is a RegEx which means $a(b|\epsilon)c = \{abc, ac\}$
- **Character Classes:** Sometimes it is convenient to express a lot of related sequential range of characters or symbols using some shorthand notations. The most common such notations even have some common names in practice. Few of them follows-
  - **Digit class:** This is written as $[0 - 9]$ which is the shorthand for the regular expression- $0|1|2|3|4|5|6|7|8|9$**.** This is often denoted & replaced by the keyword **Digit**. This refers to *any single digit from 0 to 9*.
    Note: Conventionally, a square bracket [ ] means just **any single symbol** picked from what appears inside the bracket.
  - **Alphabet class:** This is written as $[a - zA - Z]$ which is the shorthand for the regular expression- $a|b| \dots |y|z|A|B| \dots |Y|Z$. This is often denoted & replaced by the keyword **Alphabet**. This refers to *any single character from either a to z (small letters English) from A to Z (capital letters in English)*.